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DETAIL SPECIFICATION

AUDIBLE SIGNALS: ALARMS, BELLS, BUZZERS, HORNS, AND SIRENS, ELECTRONIC, SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers various types of audible signals for alarm and signaling purposes (bells, buzzers, horns, sirens, and electronic) for Naval shipboard use.

1.2 Classification. Audible signals are of the following types, characteristics, currents, and voltages, as specified (see 6.2).

1.2.1 Basic indicator. The basic indicator for naval interior communication equipment is "IC/".

1.2.2 General type. General type of signal is designated by a single letter as follows:

Letter	Type of signal
B	Bell (see 6.11.1)
Z	Buzzer (see 6.11.2)
H	Horn (see 6.11.4)
S	Siren (see 6.11.5)
E	Electronic (see 6.11.3)

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1.2.3 Special characteristics. Special characteristics of the general type are designated by a number as follows:

Special characteristics		
Bells	**Number**	**Characteristic**
	1	3-inch diameter gong
	2	8-inch diameter gong
	3	Cowbell-shaped gong
	4	Single stroke bar chime
	5	Bus failure alarm system
Buzzers	**Number**	**Characteristic**
	1	Low intensity, watertight
Horns [1/]	**Number**	**Characteristic**
	1	Non-resonated, 100 to 600 Hertz (Hz), 95 decibels (dB)
	2	Resonated, 250 to 350 Hz, 95 dB
	3	Resonated, 300 to 350 Hz, 112 dB (general and motor boat navigational signal)
	6	Mechanical, 100 to 600 Hz, motorboat use, 90 dB
	7	Mechanical, 200 to 350 Hz, fog signal, 110 dB
	8	Motor-operated, 100 to 600 Hz, 103 dB
Sirens	**Number**	**Characteristic**
	1	Low pitch, 600 to 1,200 Hz
	2	High pitch, 1,750 to 2,500 Hz
Electronic signals	**Number**	**Characteristic**
	1	Bus failure alarm signal 800 to 1,800 Hz
	3	Switchboard, audible, and remote signals 500 to 1,000 Hz

NOTE:

[1/] Navigational horns constructed of corrosion-resistant materials will be identified by the letter “N” as the last character in the type designation.

1.2.4 Supply current. Type of supply current is designated by a single letter as follows:

Letter	Type of supply current
D	Direct current (DC)
S	60 Hz alternating current (AC)
F	400 Hz AC

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1.2.5 Supply voltage. Supply voltage is designated by a number as follows:

Number	Supply voltage
1	6-volt (V) supply
2	12-V supply
3	24-V supply
4	115-V supply
5	440-V supply

1.3 Part or identifying number (PIN). PINs to be used for the audible signals furnished under this specification are created as follows:

M	**15303**	**-**	**IC/**	**X**	**X**	**X**	**X**
Prefix for Military Specification	Specification number		Basic indicator for naval interior communication equipment	General type of signal (see 1.2.2)	Special characteristic of the general types (see 1.2.3)	Type of supply current (see (1.2.4)	Supply voltage (see 1.2.5)
Example: The PIN for a bell with a direct current, 6-volt supply, and 3-inch diameter gong would be IC/B1D1.							

1.3.1 Optional part identification suffix. Additional characters may follow the end of a designator for the contractor to distinguish differing unit components or visual characteristics that do not alter the form, fit, or function of the end unit as described herein.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

MIL-A-17196 - Alarm Switchboards, Alarm Panels, Remote Sentry Alarm Switch, Signal Units, and Special Alarm Connection Box, Naval Shipboard

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

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MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-690	-	Failure Rate Sampling Plans and Procedures
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1310	-	Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
MIL-STD-1474	-	Noise Limits

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ANSI/ASA S12.55	-	Acoustics – Determination of Sound Power Levels and Sound Energy Levels of Noise Sources Using Sound Pressure – Precision Methods for Anechoic and Hemi-Anechoic Rooms

(Copies of this document are available online at https://asastandards.org/.)

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM B209 - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM B545 - Standard Specification for Electrodeposited Coatings of Tin

(Copies of these documents are available online at www.astm.org.)

INSTITUTE OF PRINTED CIRCUITS (IPC)

IPC J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies

IPC J-STD-004 - Requirements for Soldering Fluxes

IPC J-STD-006 - Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications

(Copies of these documents are available online at www.ipc.org.)

SAE INTERNATIONAL

SAE-AMS-A-21180 - Aluminum-Alloy Castings, High Strength

SAE-J517 - Hydraulic Hose

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. However, nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Audible signals furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 New technology. Audible signals manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained. The equipment manufactured incorporating new technology must meet or exceed the requirements of this document (see 6.6).

3.3 Construction and material.

3.3.1 Metal parts. All metal parts, other than current-carrying parts, shall be of corrosion-resistant material or shall be suitably protected to resist corrosion.

3.3.1.1 Aluminum. Aluminum used for construction shall conform to alloy 2024, temper T4 of ASTM B209; alloy 5052, temper T4 of ASTM B209; or other materials specifically approved by the procuring activity (see 6.6).

3.3.1.2 Dissimilar metals. Selection of metals for use in electronic equipment should be made in accordance with the requirements of MIL-STD-889. Where the requirements herein preclude the insulation of incompatible metal combinations as identified in MIL-STD-889 from one another, the metal combination shall be isolated by plating or coating, and the contact surface shall be insulated from exterior environments.

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3.3.1.3 Fasteners. Fastener hardware shall be of material that is anti-corrosive or austenitic corrosion-resistant steel and to commercial specification. No finished plating is required on stainless steel.

3.3.2 Mounting. Unless otherwise specified herein (see 3.10.1.1 and 6.2), all signals shall be two- or three-point vertically mounted.

3.3.3 Adjustability. Audible signals shall provide for adjustment for optimum audible output. Adjustment features shall be constructed to be securely locked to ensure stability of positioning.

3.3.4 Painting. Audible signals shall be painted; however, the surfaces of audible signals' identification plates, bell gong, and strikers shall not be painted. Painting shall not be required for the interior and exterior of an audible signal's brass enclosure or brass box assembly.

3.3.5 Grounding and bonding. Short circuits or grounds that could be caused by excessive motion of armatures or other moving parts shall be prevented. Bonding shall be in accordance with MIL-STD-1310.

3.3.6 Shielding and electromagnetic interference reduction. Shielding and electromagnetic interference characteristics shall be in accordance with MIL-STD-461 for all types of audible signals. Radio Frequency Interference (RFI) filters in compliance with MIL-STD-1399-300-1 shall be used when necessary to enable the audible signals to meet these shielding and electromagnetic interference characteristics.

3.3.7 Special tools. Audible signals shall be constructed so that special tools will not be required for disassembly, repair, or reassembly. Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

3.3.8 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.9 Toxicity and prohibited materials.

3.3.9.1 Toxicity. When evaluated in accordance with 4.4.12.1, the signals shall pose no serious or high risk to the health of personnel or the environment when used for their intended purpose (see 4.4.12.1 and 6.10).

3.3.9.2 Prohibited materials. The signals shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

3.3.10 Solder connections. The soldered electrical connections shall be manufactured and inspected in accordance with class 2 or class 3 guidelines of IPC J-STD-001. Solder and solder flux shall be in accordance with IPC J-STD-004 and IPC J-STD-006, respectively.

3.3.11 Pure tin. The use of pure tin as an underplate or final finish is prohibited both internally and externally. Tin content of audible signal components and solder shall not exceed 97 percent by mass. Tin shall be alloyed with a minimum of 3 percent lead by mass. Refer to ASTM B545 for additional information.

3.4 Power supply tolerances. The audible signals shall pass the tests of 4.4.2 and 4.4.3 when operated from a power supply at the upper and lower voltage and frequency limits specified in table I.

3.4.1 Insulation resistance. Insulation resistance shall not be less than 10 megohms at 500 volts direct current (Vdc), at approximately room temperature, and at approximately 50 percent relative humidity.

3.4.2 Dielectric strength. The electrical components shall be able to withstand a minimum voltage level of ten times the input power voltage for a duration of no less than 60 seconds. The components shall show no evidence of malfunction when tested in accordance with 4.4.15..

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TABLE I. Power supply voltage and frequency limits.

Normal power source		Upper limit		Lower limit	
6	Vdc	7.1	V	5.25	V
12	Vdc	14.2	V	10.5	V
24	Vdc	28.4	V	21	V
115	Vdc	112	V	108	V
115	V	123	V	107	V
60	Hz	63	Hz	57	Hz
115	V	123	V	107	V
400	Hz	420	Hz	380	Hz
440	V	471	V	410	V
60	Hz	63	Hz	57	Hz
440	V	471	V	410	V
400	Hz	420	Hz	380	Hz

3.5 Types of audible signals. The various types of audible signals shall be in accordance with table II. For those signals designated to be resonated, the signal shall be fitted with a resonating projector or chamber so as to produce a tone containing principally the fundamental frequency and its harmonics.

TABLE II. Types of audible signals.

Type IC/	Acoustical output		Degree of enclosure	Figure number
	Fundamental frequency Hz	dB (minimum)		
B1D1	----	73	Drip proof 3/	4
B1D2	----	73	Drip proof 3/	4
B1D3	----	73	Watertight 1/	4
B1D4	----	73	Drip proof 3/	4
B1S4	----	75	Watertight 1/	4
B2D1	----	85	Drip proof 3/	2
B2D2	----	85	Drip proof 3/	2
B2D3	----	85	Drip proof 3/	2
B2D4	----	85	Drip proof 3/	2
B2S4	----	85	Watertight 1/	2
B2S4 (EXP)	----	85	Watertight 1/ Explosion proof 2/	2
B3D2	----	85	Drip proof 3/	3
B3D3	----	85	Watertight 1/	3
B3D4	----	85	Drip proof 3/	3
B3S4	----	85	Watertight 1/	3

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TABLE II. Types of audible signals – Continued.

Type IC/	Acoustical output		Degree of enclosure	Figure number
	Fundamental frequency Hz	dB (minimum)		
B5S5	----	78	Drip proof 3/	1
Z1D1	----	68	Watertight 4/	5
Z1D2	----	68	Watertight 1/	5
Z1D3	----	68	Watertight 1/	5
Z1D4	----	73	Watertight 1/	5
Z1S4	----	73	Watertight 1/	5
Z1F4	----	73	Watertight 1/	5
H1D4	100 to 600	93	Watertight 1/	9
H1S4	100 to 600	93	Watertight1/	10
H2D4	250 to 350 resonated	93	Watertight 1/	8
H2S4	250 to 350 resonated	93	Watertight 1/	8
H3D2	300 to 350 resonated	110	Watertight 1/	7
H3D2N	300 to 350 resonated	110	Watertight 1/	7
H3D3	300 to 350 resonated	110	Watertight 1/	7
H3D3N	300 to 350 resonated	110	Watertight 1/	7
H3D4	300 to 350	110	Watertight 1/	7
H3D4N	300 to 350	110	Watertight 1/	7
H3S4	300 to 350 resonated	110	Watertight 1/	7
H3S4N	300 to 350 resonated	110	Watertight 1/	7
H6	100 to 600	88	4/	----
H6N	100 to 600	88	4/	----
H7	200 to 350	108	4/	----
H7N	200 to 350	108	4/	----
H8D3	100 to 600	101	Watertight 1/	6
H8D4	100 to 600	101	Watertight 1/	6
H8S4	100 to 600	101	Watertight 1/	6
H8F4	100 to 600	101	Watertight 1/	6
S1D4	600 to 1,200	103	Drip proof 3/	11
S1S4	600 to 1,200	103	Drip proof 3/	11

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TABLE II. Types of audible signals – Continued.

Type IC/	Acoustical output		Degree of enclosure	Figure number
	Fundamental frequency Hz	dB (minimum)		
S1S4CMS	600 to 1,200	103	Drip proof 3/	11
S2D4	1750 to 2,500	103	Drip proof 3/	11
S2S4	1750 to 2,500	103	Drip proof 3/	11
S2S4CMS	1750 to 2,500	103	Drip proof 3/	11
E1D1	800 to 1,800	88	Drip proof 3/	12
E1D2	800 to 1,800 Horn tone	88	Drip proof 3/	12
E3D2	500 to 1,000 variable	88	-----	----

NOTES:

1/ Watertight 3 feet, submerged (using MIL-STD-108).

2/ Explosion proof against highly volatile petroleum vapor (see 3.7.2.1).

3/ Drip proof in accordance with MIL-DTL-2036.

4/ Watertight 3 feet, submerged (using MIL-STD-108). Pump, if used, shall be drip proof.

3.5.1 Enclosure. The audible signals shall be in accordance with the enclosure requirements of MIL-DTL-2036. The degree of enclosure shall be as shown in table II.

3.6 Acoustical outputs.

3.6.1 Output level. Audible signals shall meet the minimum dB output specified in table II. Sound pressure output tests shall be based on a reference level of 20 micropascals (µPa) and shall be measured at a distance of 10 feet from the signal with the microphone positioned to receive maximum output from the signal.

3.6.2 Output frequency. The fundamental frequency of the acoustical output shall be within the frequency (Hz) limits specified in table II.

3.7 Bells and buzzers.

3.7.1 Type B5S5 bell. Type B5S5 bells shall be self-actuated upon the loss of power on the 440 V bus connected to the signal's bus failure circuit to show a visual signal as well as sound an audible alarm meeting the acoustical output specified in table II. The gong striker shall be operated by an electrically-wound spring that shall continuously operate the alarm for a minimum of 15 to 45 seconds after a bus power failure. The power supply for the winding circuit shall be 115 Volts AC (Vac), 60 Hz with operating tolerance in accordance with 3.4. The visual signal shall be a rotary drum or disk type annunciator. The visual and audible alarms shall be actuated by a relay that is continuously energized from the bus whose failure is to be indicated. The annunciator shall show a white field to indicate that the bus is energized and a red field to indicate that the bus supply is or has been off (see 3.7.1.2). Vibrations shall not cause the alarm target to move or give false indications.

3.7.1.1 Relay operational voltage. The relay for the type B5S5 bell shall be energized by 440 Vac, 60 Hz. The alarm shall operate with the input voltage reduced to 350±20 V. After the input voltage drops below 350±20 V the relay shall not reset until the input voltage is restored to a minimum of 400±20 V.

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3.7.1.2 Signal operations. When the bus supply is restored, the relay shall reset, the alarm gong shall cease ringing (if still operating) and the spring winding circuit shall be restored, but the annunciator shall continue to show the red field until manually reset. After the spring has been fully wound, it shall remain in the wound condition until failure of the bus supply causes its operation, even though the winding power supply is disconnected. Time required to wind the spring shall not exceed 2 minutes. These signals shall not be inclined more than 45 degrees from their normal vertical mounting position for specified output. A pushbutton shall be provided for simulating a bus failure condition.

3.7.1.3 Assembly and mounting dimensions. Type B5S5 bells shall meet the assembly and mounting dimensions shown on figure 1.

3.7.2 Type B2D1-4, B2S4, and B2S4 (EXP) bells. These 8-inch diameter gong bells shall meet the assembly and mounting dimensions shown on figure 2.

3.7.2.1 Type IC/B2S4 (EXP) bell. In addition to meeting watertight requirements in accordance with MIL-STD-108, this bell shall also meet the explosion proof requirements of MIL-DTL-2036 against higher volatile petroleum vapor.

3.7.3 Type B3D2-4 and B3S4 bells. These cowbell shaped gong bells shall meet assembly and mounting dimensions shown on figure 3.

3.7.4 Type B1D1-4 and B1S4 bells. These 3-inch diameter gong bells shall meet the assembly and mounting dimensions shown on figure 4.

3.7.5 Type Z1D1-4, Z1S4, and Z1F4 buzzers. All buzzers shall meet the assembly and mounting dimensions shown on figure 5.

3.8 Horns.

3.8.1 "N" suffix. Horns with an "N" suffix shall be constructed of corrosion-resistant materials.

3.8.2 Type H6 and H6N horns. These audible signals are intended as an auxiliary horn for motor boats and shall be hand operated as a mechanical signal. The horn construction shall permit maintaining the signal shown in table II for a period of not less than 1-second-per-stroke. The horn shall withstand a pressure three times greater than the operating pressure without bursting. The force of each stroke required to produce the signal shall not exceed 75 pounds. The horn shall be capable of vertical or horizontal mounting without operational or functional impact.

3.8.3 Type H7 and H7N horns. These horns shall be air horns manually operated as a ship's auxiliary fog signal. The horn shall be portable and shall be mounted on a non-slip stirrup or similar non-slip pedestal. The horn construction shall permit maintaining a blast in accordance with table II for a period of not less than 12 seconds. The horn shall sound short, distinct blasts at a rate of one per second. The force required to produce the signal specified in table II shall not exceed 80 pounds on a down stroke for single action pump. The size of the horn and pedestal shall be such as to permit stowing in a box having inside dimension of 30 by 16 by 12 inches. The horn shall be equipped with a projector at least 15 inches long. Where flexible air hose is incorporated in the design, the hose shall be in accordance with SAE-J517. The horn shall withstand a pressure three times greater than the operating pressure without bursting. A pressure gauge shall be provided on any accumulator tank of the horn and a pressure relief valve shall be provided for protection. The pumping rate to sound the horn to conform to the requirements of table II continuously for a 12-second period with a 1-second interlude between 12-second periods for a minute of this type of operation shall be 30-down-strokes-per-minute, or less. A spring return button or lever type control device shall be included to control the sounding of the alarm. A condensate drain plug shall be provided in any accumulator tank.

3.8.4 Type H8D3, H8D4, H8S4, and H8F4 horns. These horns shall incorporate a motor-driven ratchet arranged to strike an anvil on the diaphragm. The acoustical output shall include a mixture of as many frequencies as practical. These H8 type horns shall meet the assembly and mounting dimensions illustrated on figure 6.

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3.8.5 Type H3D2-4 and H3S4 horns. These horns shall be mounted in a position to prevent water or moisture from standing or collecting in the projector assembly or between the diaphragm and base plate assembly. These H3 type horns shall meet the assembly and mounting dimensions illustrated on figure 7.

3.8.6 Type H2D4 and H2S4 horns. These horns shall be mounted in a position to prevent water or moisture from standing or collecting in the projector assembly. These H2-type horns shall meet the assembly and mounting dimensions illustrated on figure 8.

3.8.7 Type H1D4 horn. This horn shall be mounted in a position to prevent water or moisture from standing or collecting in the projector assembly. The H1D4-type horn shall meet the assembly and mounting dimensions illustrated on figure 9.

3.8.8 Type H1S4 horn. This horn shall be mounted in a position to prevent water or moisture from standing or collecting in the projector assembly. The H1S4-type horn shall meet the assembly and mounting dimensions illustrated on figure 10.

3.9 Sirens. The siren housing design shall ensure prevention of injury to personnel by coming in contact with any moving parts of the siren. The aluminum alloy for the construction of the siren shall be in accordance with SAE-AMS-A-21180.

3.9.1 AC supply current sirens. For siren motors in table II with a “CMS” suffix, the line-to-ground RFI filters in the siren (see 3.3.6) shall comply with MIL-STD-1399-300-1 and MIL-STD-461.

3.9.2 Assembly and mounting dimensions. Type S1D4, S2D4, S1S4, S2S4, S1S4CMS, and S2S4CMS sirens shall meet the assembly and mounting dimensions illustrated on figure 11.

3.10 Electronic signals.

3.10.1 Type E1D1 electronic audible signal. The E1D1-electronic signal shall be self-actuated upon loss of power on the 115 V bus connected to the signal terminals to show a visual signal as well as sound audible alarm meeting the acoustical output specified in table II. The unit shall consist of an electronic solid-state oscillator or tone generator arranged to operate a loudspeaker, howler, or other transducer to provide the audible signal upon loss of power on the supervised bus. Power for the electronic circuitry shall be derived from a self-contained rechargeable 225 milliampere-hour (mAh) (minimum) nickel cadmium (NICAD) battery, which shall be charged from the supervised bus at a rate not exceeding the 30-hour rate.

3.10.1.1 Signal operations. Upon loss of power on the bus, the audible signal shall sound for a minimum of 15 minutes and then shall be silenced by a manually operated pushbutton. The red field of the visual signal shall be displayed until power is restored and the reset pushbutton is operated, at which time the red field of the indicator shall disappear. A neon “power-on” light with white lens shall be connected directly across the bus supply. The front panel shall contain the transducer, power-on light, silence-reset pushbutton, and visual indicator. The unit shall not exceed 4 inches wide by 4 inches deep by 5½ inches high and shall be universal mounted. The unit shall supervise 115 Vdc, 60 or 400 Hz power without modification and without observing polarity of the DC voltage. The alarm shall sound when the supervised voltage is reduced to between 50 and 70 V. Resetting of the audible signal shall be feasible upon restoration of the voltage to between 75 and 95 V.

3.10.1.2 Type E1D2 electronic audible signal. The E1D2-electronic signal shall operate as a part of the airflow indicator and alarm system. In this application, the unit shall operate as a power failure alarm, a power supply, and an alarm signal. The unit shall be identical to the type IC/E1D1, except that it shall be powered by a 12 V rechargeable NICAD battery and shall be made safe by utilizing sealed contacts.

3.10.1.3 Assembly and mounting dimensions. Type E1D1 and E1D2-electronic audible signals shall meet the assembly and mounting dimensions illustrated on figure 12.

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3.10.2 Type E3D2 electronic audible signal. The E3D2-electronic signal shall be an integral component of the IC/SM type standard Navy alarm switchboard. The E3/D2-electronic audible signal shall produce three distinct tones having a fundamental frequency varying between 500 and 1,000 Hz, such as steady siren tone (supervisory failure), siren wail tone (alarm condition), and siren yelp (power failure alarm). Terminals shall be provided to serve as a means to interconnect each of the above alarms with its applicable relay. The unit shall be a completely transistorized NICAD battery-powered tone generator that shall sound an alarm from two speakers simultaneously in case of power failure for a continuous period of 30 minutes and shall be in accordance with the audible output requirement shown in table II. The NICAD battery shall be so connected that it will float on the line under a continuous charging rate sufficient to maintain the battery in ready condition at all times. Refer to MIL-A-17196 for form and fit requirements of the IC/E3D2 electronic signal in the IC/SM type alarm switchboard.

3.11 Temperature and humidity.

3.11.1 Humidity testing upper temperature. Audible signals shall meet the temperature and humidity parameters of MIL-STD-810 and shall meet the humidity test of MIL-STD-810 with an upper test temperature of 149±9 degrees Fahrenheit (°F) in lieu of 140±9 °F, without degradation of performance.

3.11.2 Extended storage. All audible signals shall be capable of extended storage in any ambient temperature between the limits of -40 and +167 °F. Audible signals shall meet the extended storage test specified in 4.4.9.2.

3.11.3 Extreme temperature. Watertight audible signals (see table II) shall pass the tests of 4.4.2 and 4.4.3 at normal input voltage and frequency for the temperature range of -65.2 °F to +149 °F.

3.12 Transient voltage (AC signals). Damage shall not be sustained by the AC audible signals when subjected to the spike voltage of MIL-STD-1399-300-1.

3.13 Shock and vibration. Audible signals, when subjected to the vibration test specified in 4.4.6 and shock test specified in 4.4.7, shall show no mechanical derangement or damage, or change of output characteristics in excess of 1.5 dB from the actual values originally determined for the sets, as per the acoustical output in table II.

3.14 Accelerated life. Audible signals shall meet the accelerated life test of 4.4.4.

3.15 Salt spray. Audible signals (watertight enclosure) shall show no corrosion or other damage when subjected to the salt spray test of 4.4.17.

3.16 Reliability. Each item shall meet a specified mean time between failures (MTBF) of not less than 1×10^4h (failure rate not greater than 1 failure/10^4h).

3.16.1 Reliability prediction. A reliability prediction shall be developed using the stress analysis procedures and failure rates of MIL-STD-690.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to the Naval Sea Systems Command (NAVSEA). Three audible signals of each type for which qualification is requested shall be subjected to the tests listed in table III. Tests shall be performed in the order listed.

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TABLE III. Qualification inspection.

Test	Requirement paragraph	Test reference
Workmanship/general examination 1/	3.2, 3.3.1, 3.3.2, 3.3.4, 3.3.5, and 3.3.7 through 3.3.11	4.4.13
Operation	3.3.3, 3.7, 3.8, 3.9, and 3.10	4.4.1
Acoustical output 1/	3.5 and 3.6	4.4.2 and 4.4.3
Electromagnetic interference	3.3.6	4.4.8
Transient voltage	3.12	4.4.10
Enclosure	3.2, 3.3.1, and 3.5.1	4.4.16
Salt spray 2/	3.15	4.4.17
Extended storage	3.11.2	4.4.9.2
Humidity	3.11.1	4.4.9.1
Extreme temperature for exposed location	3.11.3	4.4.9.3
Vibration (vital equipment)	3.13	4.4.6
Accelerated life 1/	3.14	4.4.4 and 4.4.5
Dielectric strength 1/	3.4.2	4.4.15
Insulation resistance 1/	3.4.1	4.4.14
Shock	3.16	4.4.7
Explosion (type IC/B2S4 [EXP] only)	3.7.2.1	4.4.11

NOTES:

1/ Tests required for retention testing.

2/ Applies to watertight signals only.

4.2.1 Extension of qualification.

4.2.1.1 DC bells and buzzers. Tests conducted on any one of the DC bells listed will be extended to qualify all of the other bells shown. Testing any one of the DC buzzers shown will qualify all of the other buzzers shown. Successful testing of any one of the DC bells does not qualify any of the DC buzzers; conversely, successful testing of any one buzzer does not qualify any of the bells.

IC/B1D1, IC/B1D2, IC/B1D3, IC/B1D4,
IC/B2D1, IC/B2D2, IC/B2D3, IC/B2D4,
IC/B3D2, IC/B3D3, IC/B3D4,
IC/Z1D1, IC/Z1D2, IC/Z1D3, IC/Z1D4

4.2.1.2 AC bells, AC buzzers, horns, sirens, and electronic signals. Grouping of units for qualification testing shall be in accordance with table IV. Approval of the test unit shown in column 1 will extend approval to the units shown in column 2 of the same category, provided additional testing is done as required by column 3.

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TABLE IV. Qualification inspection family grouping.

Column 1	Column 2	Column 3
Test unit (approved type) (type IC/)	**Family group (extended approval to) (type IC/)**	**Except for test**
B2S4 (EXP)	B2S4 (EXP), B2S4, and B3S4	
B1S4	B1S4	
B5S5	B5S5	
Z1S4	Z1S4 and Z1F4	
H1D4	H1D4 and H1S4	
H2S4	H2S4 and H2D4	
H3S4N	H3D2 and H3D2N H3D3 and H3D3N H3D4 and H3D4N H3S4 and H3S4N	
H6N	H6 and H6N	
H7N	H7 and H7N	
H8S4	H8D3, H8D4, H8S4, and H8F4	
S1D4	S1D4, S1S4, and S1S4CMS	
S2D4	S2D4, S2S4, and S2S4CMS	
E1D1	E1D1 and E1D2	Shock test [1/]
E3D2	E3D2	
Any DC bell listed in 4.2.1.1	All DC bells listed in 4.2.1.1	
Any DC buzzer listed in 4.2.1.1	All DC buzzers listed in 4.2.1.1	

NOTE:

[1/] A manufacturer who has successfully completed qualification tests on type IC/E1D1 may also be qualified for type IC/E1D2 upon successful completion of the shock test only on the IC/E1D2 unit (see 4.4.7).

4.2.2 Retention of qualification. To maintain qualification, one representative sample of each qualified audible signal test unit specified in table IV, column 1, from current production shall be subjected to retention inspection (see table III) at the end of every 3 years after initial qualification or a previous retention inspection. Retention testing shall be at the rated voltage and frequency captured in table I and meet the acoustical output specified in 3.5.

4.2.2.1 Rejection. If a sample tested in accordance with 4.4.2 fails to meet the requirements, at minimum, that sample type unit is subject to repeat the failed retention test, including any prior retention test(s) that the failed test is dependent upon. If the sample type unit fails the same retention test twice, then the failed unit type shall require requalification in accordance with 4.2. If the failed sample is physically modified to correct the retention failure, then the failed unit type shall require requalification in accordance with 4.2.

4.2.2.2 Change approval. A change in material, production processes, or production equipment used in the manufacture of a qualified item shall require written approval from NAVSEA. Incorporation of unapproved changes shall require requalification of the item in accordance with 4.2.

4.3 Conformance inspection. Conformance inspection shall be in accordance with table V. Tests shall be performed in the order listed.

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TABLE V. Conformance inspection.

Test	Requirement paragraph	Test reference
Group A		
Workmanship/general examination	3.2, 3.3.1, 3.3.2, 3.3.4, 3.3.5, and 3.3.7 through 3.3.11	4.3.1
Group B		
Operation	3.3.3, 3.7, 3.8, 3.9, and 3.10	4.4.1
Dielectric strength	3.4.2	4.4.15
Insulation resistance	3.4.1	4.4.14
Acoustical output (at normal voltage and frequency)	3.5 and 3.6	4.4.2 and 4.4.3

4.3.1 Sampling for conformance inspection.

4.3.1.1 Lot. All audible signals of the same type presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity, or it may be the production of any convenient time period. The acceptance and rejection requirements of ANSI/ASQ Z1.4 shall apply.

4.3.1.2 Sampling for group A examination. A random sample of audible signals shall be selected from each lot in accordance with table VI and shall be subjected to the group A tests in accordance with 4.3. The results of the examination shall be compared with the requirements of this specification. The acceptance and rejection criteria shall be as specified (see 6.2).

TABLE VI. Sampling for group A examination.

Number of audible signals in lot	Number of audible signals in sample
7 and under	All
8 to 15	7
16 to 40	10
41 to 110	15
111 to 300	25
301 to 500	35
501 and over	50

4.3.1.3 Sampling for group B tests (except acoustical output test for bells, buzzers, and IC/E1D1 electronic signals. All audible signals shall be subjected to the group B tests in accordance with 4.3, except for the acoustical output test for bells, buzzers, and type IC/E1D1 electronic signals (see 4.3.1.3.1). Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. Failure of any audible signal in the sample to pass the examination specified in 4.3.1.2 should be considered as a cause for rejection of the lot.

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4.3.1.3.1 Sampling for acoustical output test for bells, buzzers, and type IC/E1D1 electronic signals. Two units or 4 percent, whichever is greater, of each production lot of bells, buzzers, and type IC/E1D1 electronic signals shall be subjected to the acoustical output test of 4.4.2 and 4.4.3. The samples shall be selected after all units of the production lot have been adjusted by the manufacturer for maximum output prior to being tested for acoustical output. If any sample unit fails the acoustical output test, the lot represented by the sample shall be rejected.

4.4 Tests.

4.4.1 Design operation. The audible signals shall be examined and tested for conformance to the requirements of 3.3.3, 3.7, 3.8, 3.9, and 3.10.

4.4.2 Acoustical output. Sound pressure output tests shall be based on a reference level of 20 µPa and shall be measured at a distance of 10 feet from the signal with the microphone positioned to receive maximum output when the signal is mounted in the normal position. Measurements shall be made with an apparatus having a flat frequency response characteristic. Test shall be conducted to determine that the sound pressure output meets the requirements shown in table II (see 3.5), under the conditions of supply voltage and frequencies specified in 3.5. Measurement shall be made with instruments described and calibrated in accordance with MIL-STD-740-2 and MIL-STD-1474. Sound measurements shall be conducted in an anechoic chamber that conforms to ANSI/ASA S12.55. Calibration shall be made at 250 Hz and at sound pressure of 124±2 dB. The microphone shall be mounted so that the sound arrives at 0 degrees (perpendicular incidence) when sound measurements are made.

4.4.3 Frequency. An analysis shall be made of the acoustical signal to determine conformance to the frequency specified in table II (see 3.5 and 3.6).

4.4.4 Accelerated life.

4.4.4.1 Testing time and temperatures. All audible signals except types B5S5, H6, H6N, H7, H7N, and the IC/E electronic types shall operate continually at 30 seconds on and 60 seconds off for periods of time and under conditions of temperature and angles of tilt in accordance with table VII. The periods of time shall be measured from the time the temperature of the audible signal is stabilized at the test temperature. The operation of the audible signal shall be discontinued during the transition time to achieve the test temperature.

TABLE VII. Accelerated life testing time and temperatures.

Hours	Ambient temperature during test	Angle of tilt
8	+77 °F nominal (room temperature)	Normal mounting
8	-20.2 °F	Normal mounting
8	+149 °F	Normal mounting
3	+77 °F nominal (room temperature)	Inclined 60 degrees forward from normal
2	+77 °F nominal (room temperature)	Inclined 60 degrees back from normal
2	+77 °F nominal (room temperature)	Inclined 60 degrees right from normal
2	+77 °F nominal (room temperature)	Inclined 60 degrees left from normal

4.4.4.2 Type B5S5 bell. Type B5S5 bus failure alarm bells shall be tested in accordance with 4.4.4.1, except as follows:

a. The spring winding circuit of the signal shall be continuously energized at its normal rated voltage and frequency.

b. The bus failure circuit (protected bus) terminals shall be alternately energized and deenergized every 2 minutes. The annunciator flag shall be reset during each energized period. For the type B5S5 bell, the bus voltage shall be 440 V, 60 Hz AC.

c. The angle of tilt shall be not more than 45 degrees from the vertical mounting position.

d. In lieu of -20.2 °F, test shall be made at +41 °F.

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4.4.4.3 Type H6 and H6N horns. Type H6 and H6N horns shall be mechanically actuated for 1 second in each 5 seconds for an uninterrupted period of 48 hours.

4.4.4.4 Type H7 and H7N horns. Type H7 and H7N fog horns shall be mechanically actuated for 15 minutes each 30 minutes for an uninterrupted period of 48 hours.

4.4.4.5 Special testing conditions. All signals shall continue to operate without adjustment during the accelerated life test, except motor operated horns, for which the ratchet may be adjusted a maximum of four times during the test. Measurements of temperature rise shall be made on electrical apparatus during the life tests. The maximum temperature of coil windings during the endurance run at an ambient temperature of +149 °F (+113 °F for E1D1 and E1D2 electronic signals with NICAD batteries) shall not exceed the allowable temperature of coil insulation. The resistance method of measuring temperature rise shall be used.

4.4.4.6 Type E1D1 bus failure signals and E1D2 electronic audible signals. These units shall be tested in accordance with 4.4.4.1, except as follows:

a. The signal shall be continuously energized, with the supply varied hourly between 115 Vdc, 115 V, 60 Hz and 115 V, 400 Hz. Once every half hour the supply (protected bus) shall be de-energized for 1 minute, the audible signal silenced, the supply restored and the visual signal reset.

b. Upon completion of the 33-hour life test and retest of acoustical output, a special test shall be performed. The unit shall be operated 1,000 times without malfunction using the following procedure each time:

(1) de-energize the supply bus,

(2) silence the audible signal,

(3) restore the supply bus, and

(4) reset the visual signal.

The battery then shall be fully de-energized by removing the supply bus for 48 hours without silencing the audible signal. The supply bus shall then be restored and allowed to charge the battery for 8 hours. At the end of this period, the supply bus again shall be de-energized, and the acoustic output of the signal shall be measured. The acoustic output shall meet the requirements of table II.

c. In lieu of -20.2 °F, test shall be made at +41 °F.

d. In lieu of +149 °F, test shall be made at +113 °F.

4.4.4.7 Type E3D2 signal test conditions. The E3/D2 signals shall be tested as identified in table VIII.

TABLE VIII. IC/E3D2 signal test conditions.

Hours	Ambient temperature	Operating conditions
0 to 32	Room ambient (+77 °F)	Signal shall be sounded 5 minutes each hour
32 to 64	+41 °F	Signal shall be sounded 5 minutes each hour
64 to 96	+113 °F 95 percent relative humidity (R.H.) alternate 30-minute periods	Signal shall be sounded 5 minutes each hour
96 to 128 1/	Room ambient (+77 °F)	Signal shall be sounded 5 minutes each hour

NOTE:

1/ After 128 hours, complete the operation test of 4.4.1 and test of 4.4.5. No adjustments shall be permitted during the test.

4.4.5 Retest of acoustical output. Upon completion of the accelerated life test specified in 4.4.4, and without any adjustments being made, the audible signals shall pass the tests in accordance with 4.4.2 and 4.4.3 at normal voltage and frequency.

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4.4.6 Vibration. The vibration test shall be conducted in accordance with MIL-STD-167-1 for vital equipment, except that the signals shall be in the operating condition during periods of the time specified under accelerated life tests for each test condition. The signals shall continue to operate when energized and shall not operate when not energized.

4.4.7 Shock. Shock tests shall be conducted in accordance with MIL-DTL-901, except that single stroke signals shall not be energized. Single stroke signals shall be energized after each blow and shall continue to operate. Other signals shall be shocked while energized and shall continue to operate without further adjustment, but momentary cessation of operation (or operation of bus failure alarm signals) at the time of striking of a blow will be acceptable, provided the audible signal continues to operate satisfactorily during the remainder of the test and after completion of the shock tests. The criterion for "operate satisfactorily" will be the capability of the audible signal to pass the test of 4.4.2 and 4.4.3 at normal voltage and frequency.

4.4.8 Electromagnetic interference. Electromagnetic interference emission and susceptibility tests shall be performed in accordance with MIL-STD-461.

4.4.9 Temperature and humidity.

4.4.9.1 Humidity. The five cycle (120 hours) humidity test of procedure IV of MIL-STD-810 (upper temperature 149±9° F) shall be conducted on the audible signals in the manner specified in MIL-STD-810. The audible signals shall pass the tests of 4.4.2 and 4.4.3 at normal voltage and frequency at the performance test points of MIL-STD-810. For E1D1, E1D2, and E3D2-electronic audible signals, the upper temperature shall not exceed 113±9 °F.

4.4.9.2 Extended storage. The audible signals shall be subjected to the acoustical test of 4.4.2 and the frequency test of 4.4.3 at normal voltage and frequency subsequent to stabilization at room temperature.

4.4.9.3 Extreme temperature. The acoustical test of 4.4.2 and the frequency test of 4.4.3 at normal voltage and frequency shall be conducted for the temperature range of -65.2 °F to +149 °F for exposed locations. After the internal temperature of the audible signals stabilizes the audible signals shall pass these tests.

4.4.10 Transient voltage. The audible signals shall be tested for the ability to withstand, without damage, the spike voltage of MIL-STD-1399-300-1 applied a minimum of three times. The audible signals, after being subjected to the spike voltage, shall be subjected to the acoustical test of 4.4.2 and the frequency test of 4.4.3 at normal voltage and frequency.

4.4.11 Explosion test (type IC/B2S4 [EXP]). A test shall be conducted as described in MIL-DTL-2036 to determine if the type IC/B2S4 (EXP) audible signal is explosion proof (see 3.7.2.1).

4.4.12 Toxicity and prohibited materials.

4.4.12.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.3.9.1, as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the signals using data provided by the manufacturer/distributor to the NMCPHC (see 3.3.9.1 and 6.10).

4.4.12.2 Prohibited materials. Prohibited materials shall be verified for conformance to 3.3.9.2 as required by the qualifying activity.

4.4.13 General examination. A general examination of workmanship shall verify conformance to the requirements of 3.2, 3.3.1, 3.3.2, 3.3.4, 3.3.5, and 3.3.7 through 3.3.11.

4.4.14 Insulation resistance. Insulation resistance testing shall be conducted in accordance with MIL-STD-202 to ensure conformance with 3.4.1.

4.4.15 Dielectric strength. The dielectric strength of electrical components shall be tested in accordance with MIL-STD-202 to ensure conformance with 3.4.2.

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4.4.16 Enclosure. Audible signal enclosure shall be tested in accordance with MIL-DTL-2036 to ensure conformance with 3.2, 3.3.1, and 3.5.1. Following the enclosure test, the drip proof equipment shall operate properly and shall show no signs of accumulation of water, and watertight equipment shall show no entry of water greater than allowed by MIL-STD-108.

4.4.17 Salt spray. Salt spray shall be subjected to the salt spray test in ASTM B117 to ensure its conformance with 3.15.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The audible signals covered by this specification are to be used for signals and alarms on all Navy ships. Items manufactured to this revision are fully interchangeable with the items manufactured to the previous revision.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2).

c. Whether mounting for signals is other than specified (see 3.3.2).

d. Group A sampling acceptance and rejection criteria (see 4.3.1.2 and 6.4).

e. Packaging requirements (see 5.1).

f. Quantity required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 15303 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification SD-6. Copies of "Provisions Governing Qualification SD-6" may be obtained from https://quicksearch.dla.mil.

6.4 Acceptance and rejection criteria for group A conformance inspection. When specified (see 6.2), acceptance and rejection criteria for group A conformance inspection should be in accordance with table IX.

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TABLE IX. Acceptance and rejection numbers for group A examination.

Number of audible signals in lot	Number of audible signals in sample	Number of audible signals non-conforming on any group A examination	
		Acceptance number	Rejection number
7 and under	All	0	1
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

6.5 Audible signal drawings. When no standard drawing exists for a specific audible signal required to be produced, an existing standard drawing for an audible signal of the same general type (see 1.2.2) and having the same characteristics (see 1.2.3) as that of the required audible signal should be referenced for certain similar parts and configuration.

6.6 NAVSEA approval and direction. Deviations from specified materials, procedures, and requirements, and selection of specific alternative materials and procedures require NAVSEA approval or direction. Requests should include supporting documentation.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Tin whisker growth (see 3.3.11). The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM B545.

6.9 Environmentally preferable material (see 3.3.8). Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website http://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see 3.3.8).

6.10 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product (see 3.3.9.1 and 4.4.12.1). Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

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6.11 Definitions.

6.11.1 Bell. A hollow object/device, often electrically operated, usually cuplike, made of metal or other hard material that gives forth a ringing sound when struck.

6.11.2 Buzzer. An electromagnetic audio signaling device that generates a buzzing sound through the rapid vibration of an electric make and break mechanism through the use of AC or DC electrical power.

6.11.3 Electronic signals. Where signals are generated by use of electronic devices that supply a loudspeaker or other type transducer to produce an audible signal.

6.11.4 Horn. A horn shaped unit whose electromagnetic energy transferring and sound-producing device is located at its base.

6.11.5 Siren. An acoustical device, often electrically operated, that produces tones by the rapid interruption of a current in the air, steam, or fluid when driven against a rotating, perforated disk so as to produce a penetrating warning sound.

6.12 Subject term (key word) listing.

Acoustical

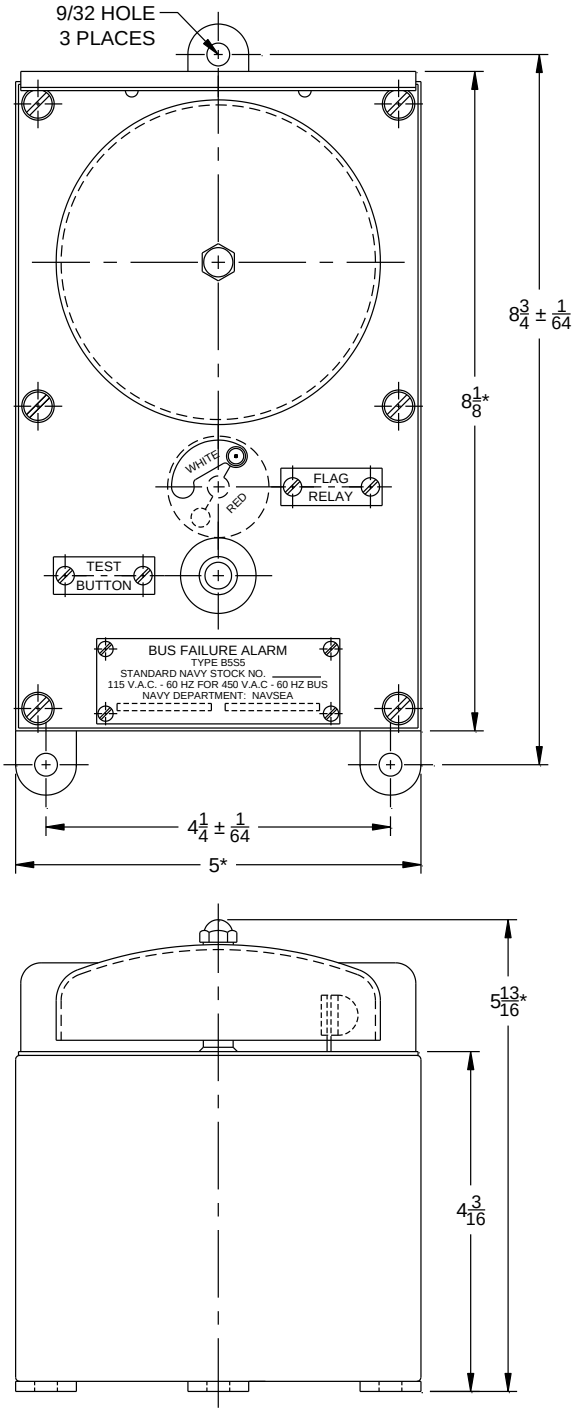
Decibels

Interior Communications

Sound

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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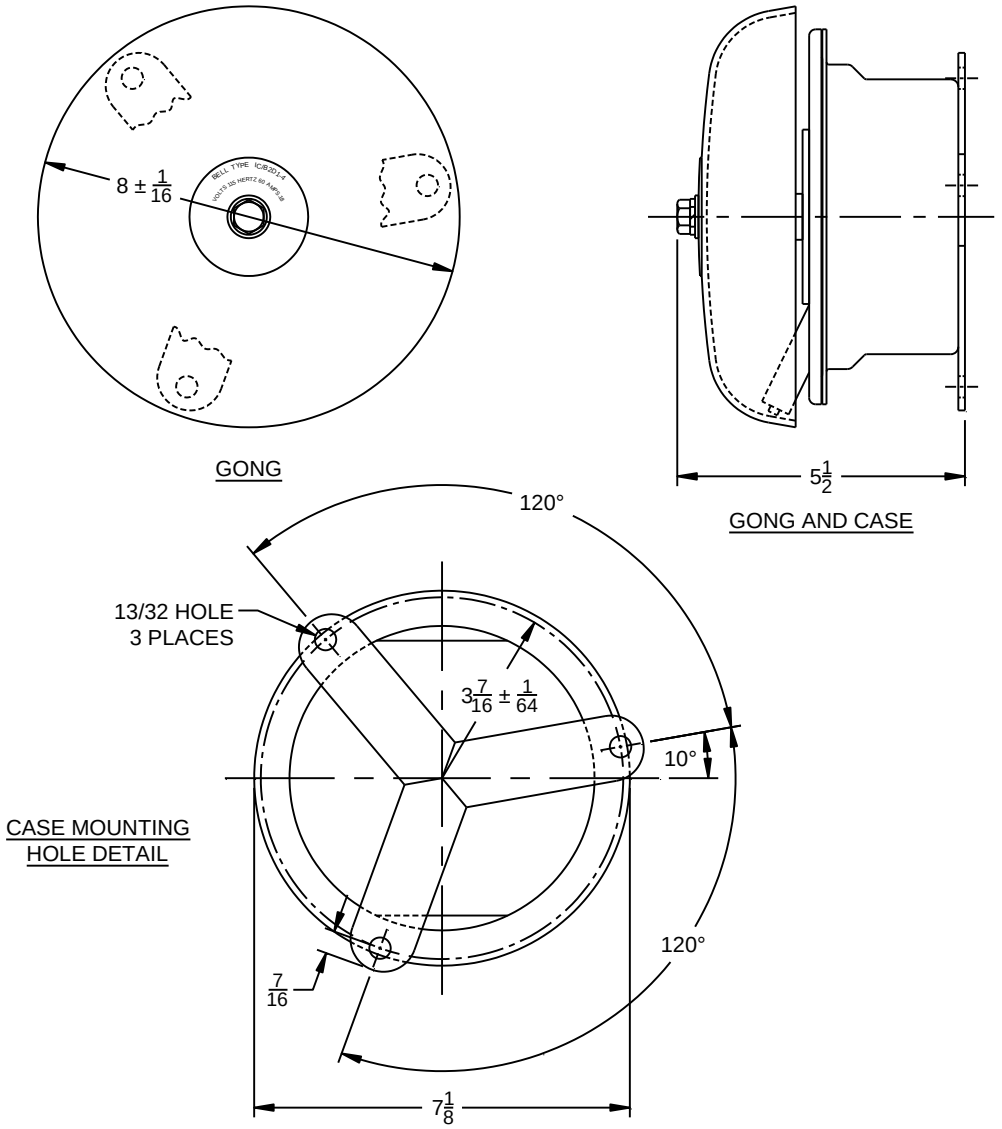


NOTES:

1. Unless specified otherwise herein, dimensions are maximum/not to exceed.
2. All dimensions are in inches.

FIGURE 1. Drip proof bell (type IC/B5S5).

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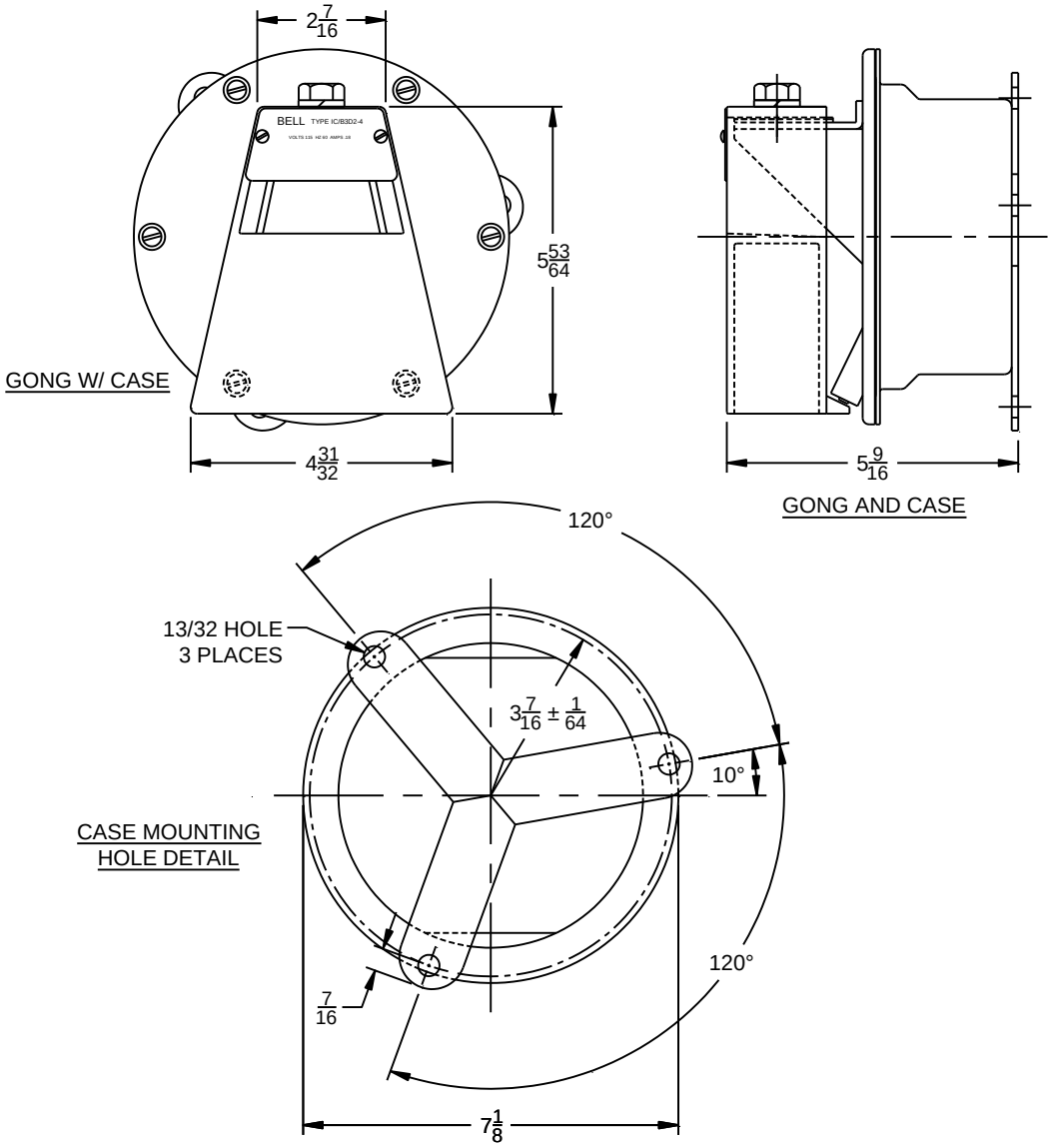


NOTES:

1. Unless specified otherwise herein, dimensions are maximum/not to exceed.
2. All dimensions are in inches.

FIGURE 2. Watertight and explosion proof bells (types IC/B2D1-4, B2S4, and B2S4[EXP]).

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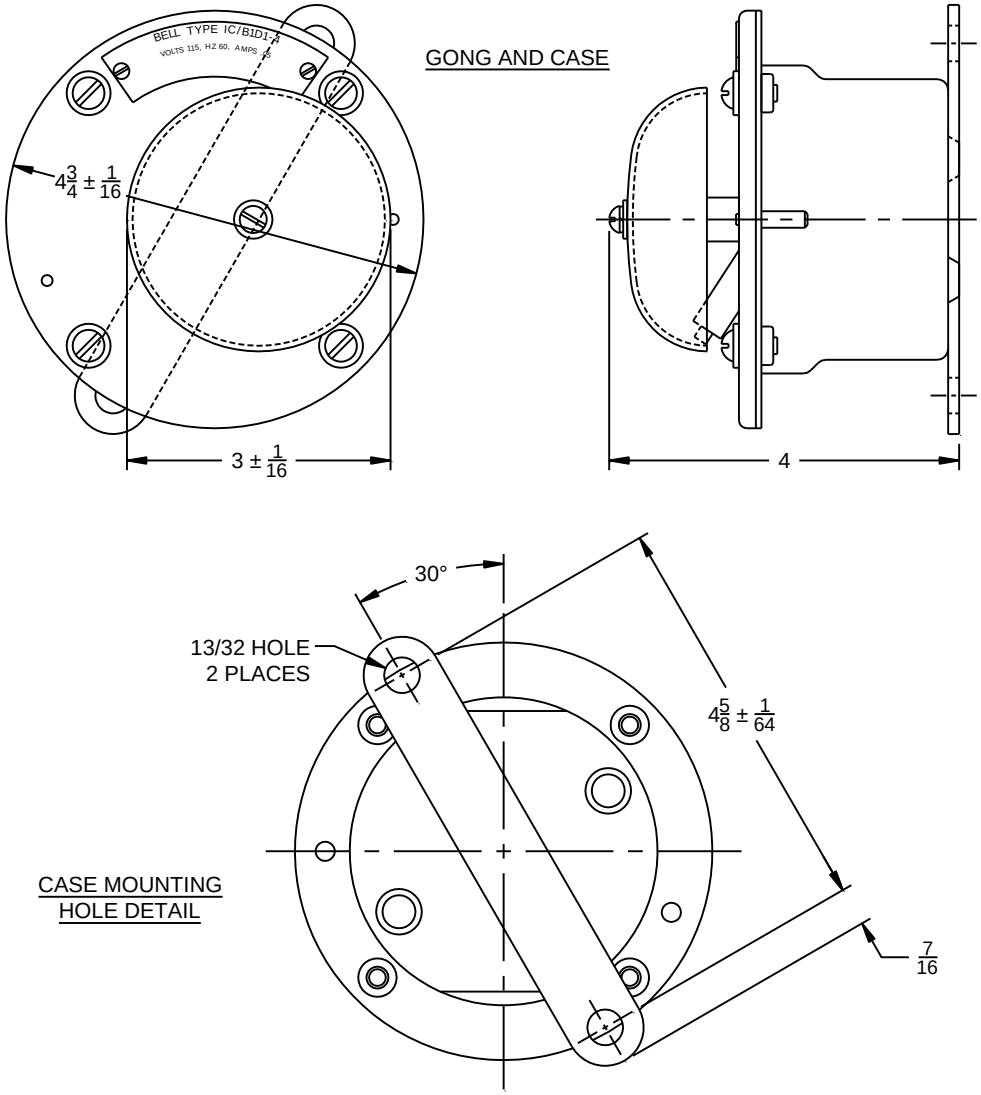


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. All dimensions are in inches.

FIGURE 3. Watertight and explosion proof bells (types IC/B3D2-4 and B3S4).

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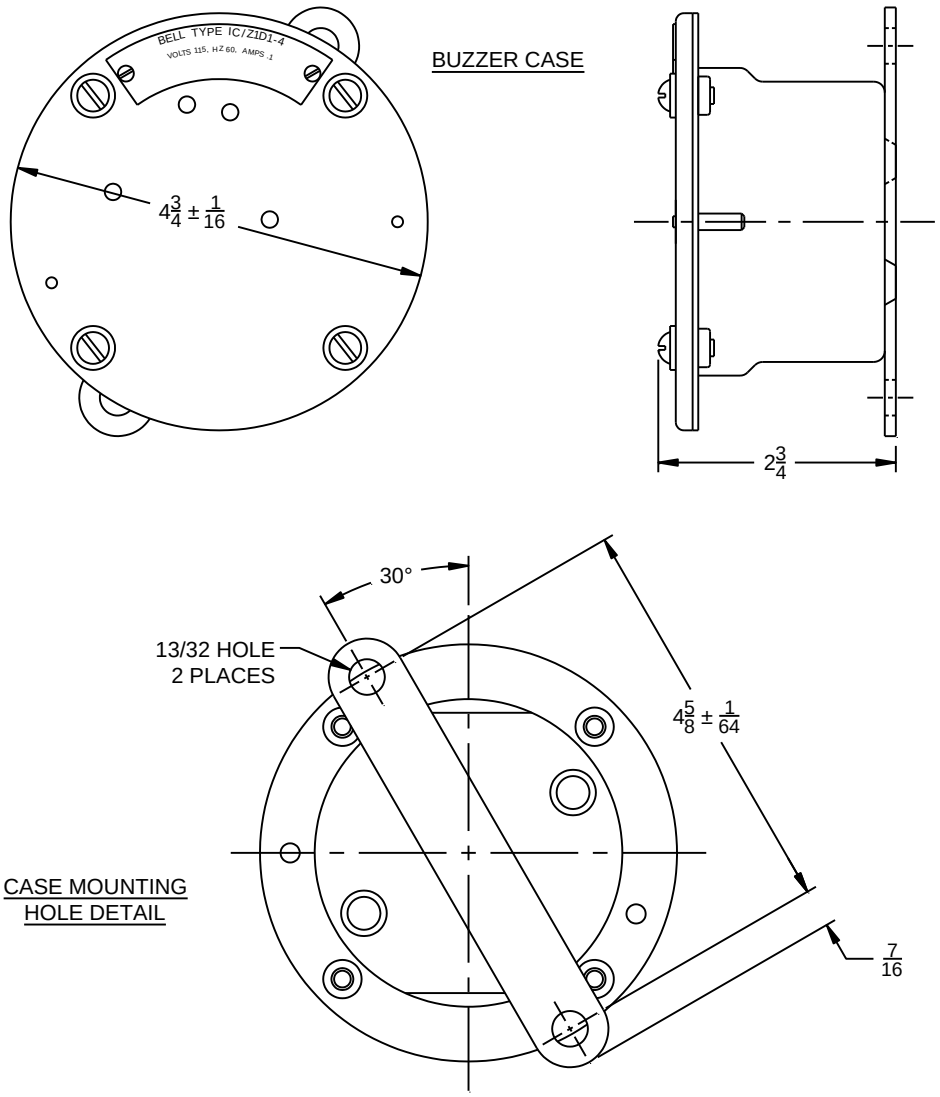


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. All dimensions are in inches.

FIGURE 4. Watertight bell (type IC/B1D1-4 and B1S4).

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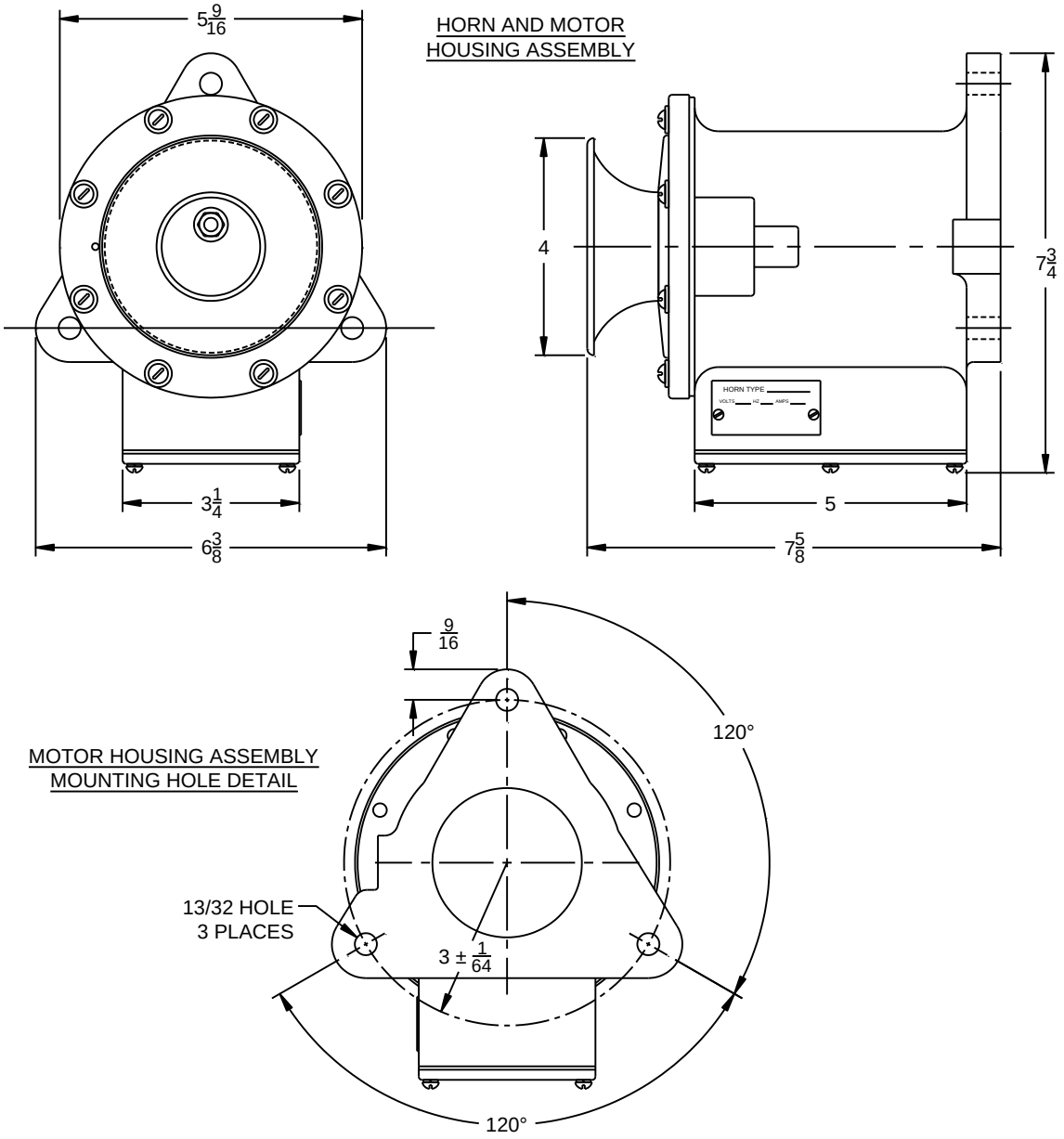


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. All dimensions are in inches.

FIGURE 5. Watertight buzzer (type IC/Z1D1-4, Z1S4, and Z1F4).

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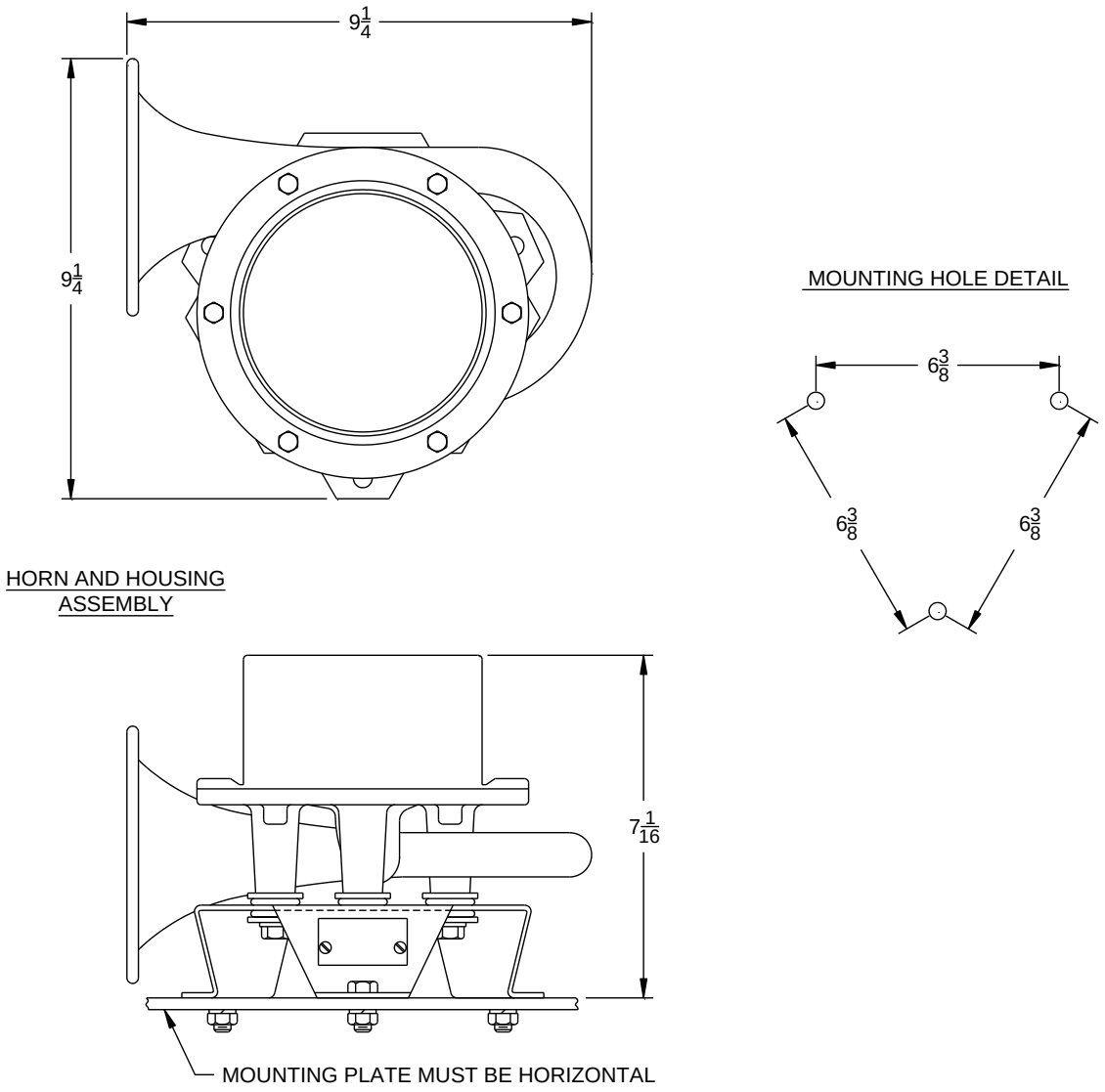


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. All dimensions are in inches.

FIGURE 6. Watertight motor-driven horns (type IC/H8D3-4, H8S4, and H8F4).

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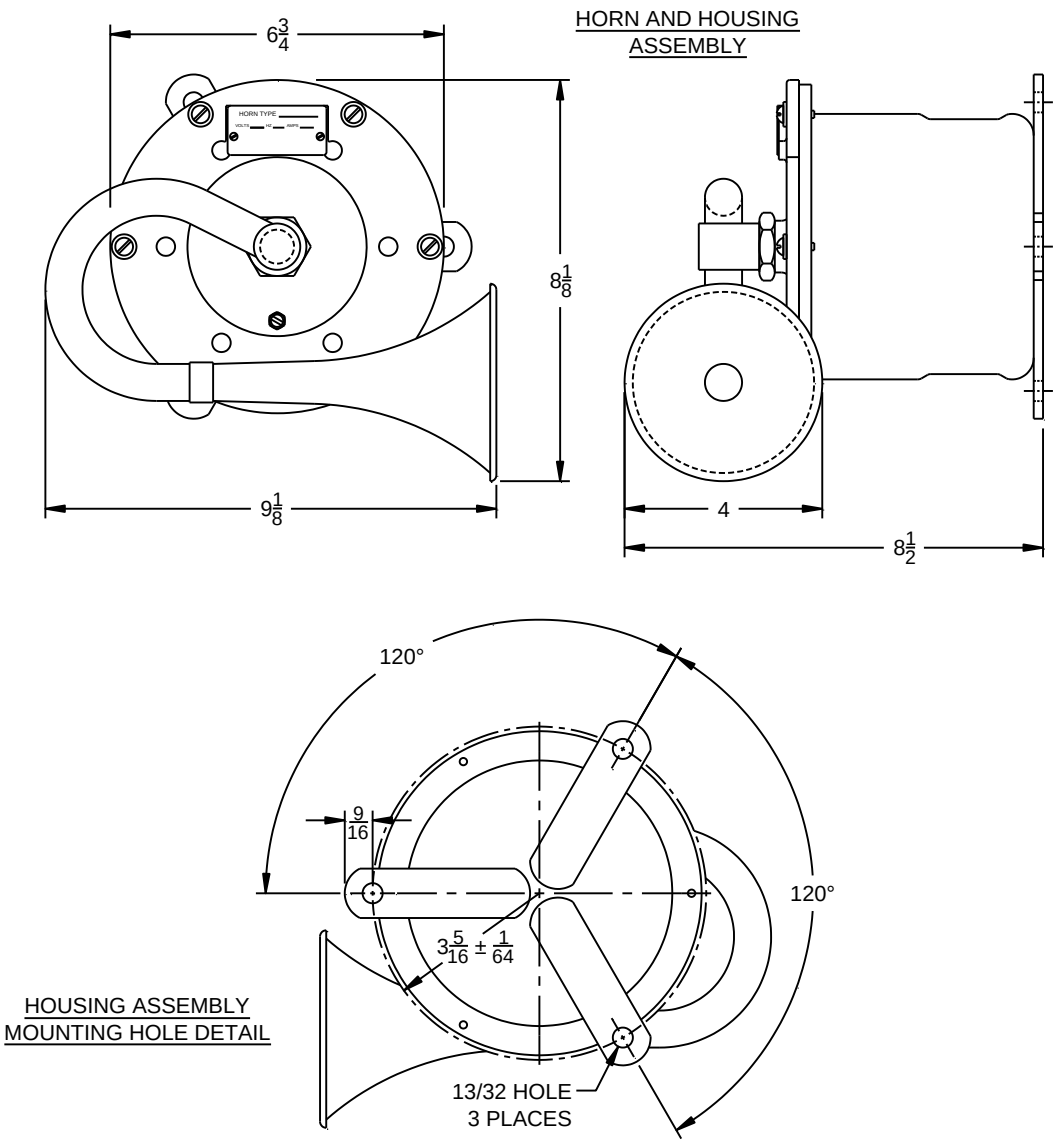


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. Horn must be mounted as shown in order to prevent water or moisture from standing in the projector assembly or between the diaphragm and the base plate. Mounting plate furnished by installing activity.
3. All dimensions are in inches.

FIGURE 7. High intensity, watertight, resonated horns (type IC/H3D2-4, H3D2N-4N, H2S4, and H3S4N).

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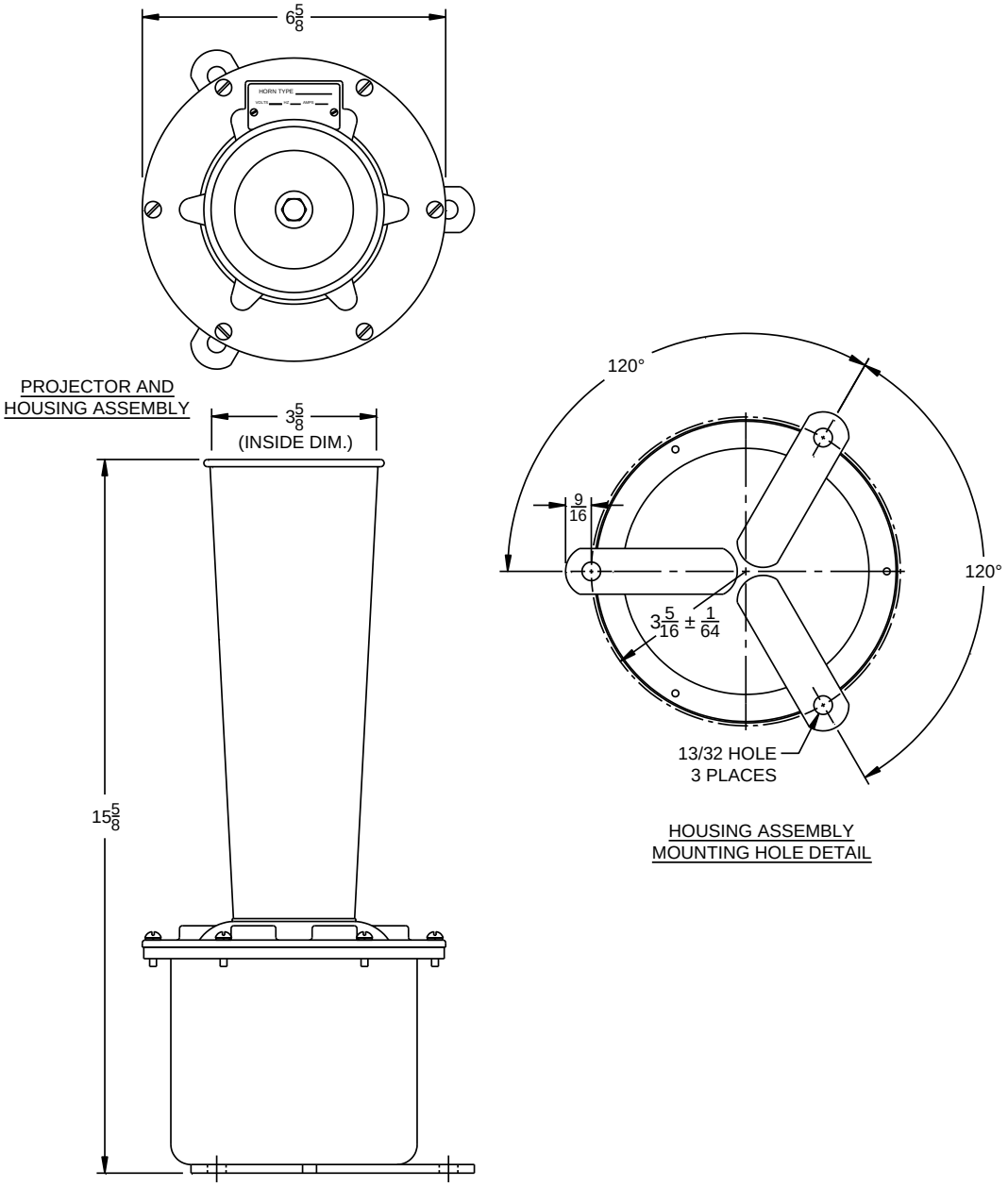


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. To obtain maximum sound output, horn should be bulkhead mounted using three spacer bosses ¾ O.D.X. ½ L with ⅜-16 tapped hole spacers to be welded to bulkhead on 6⅝-16 diameter bolt circle. Spacers furnished by installing activity.
3. All dimensions are in inches.

FIGURE 8. High intensity, watertight, resonated horns (type IC/H2D4 and H2S4).

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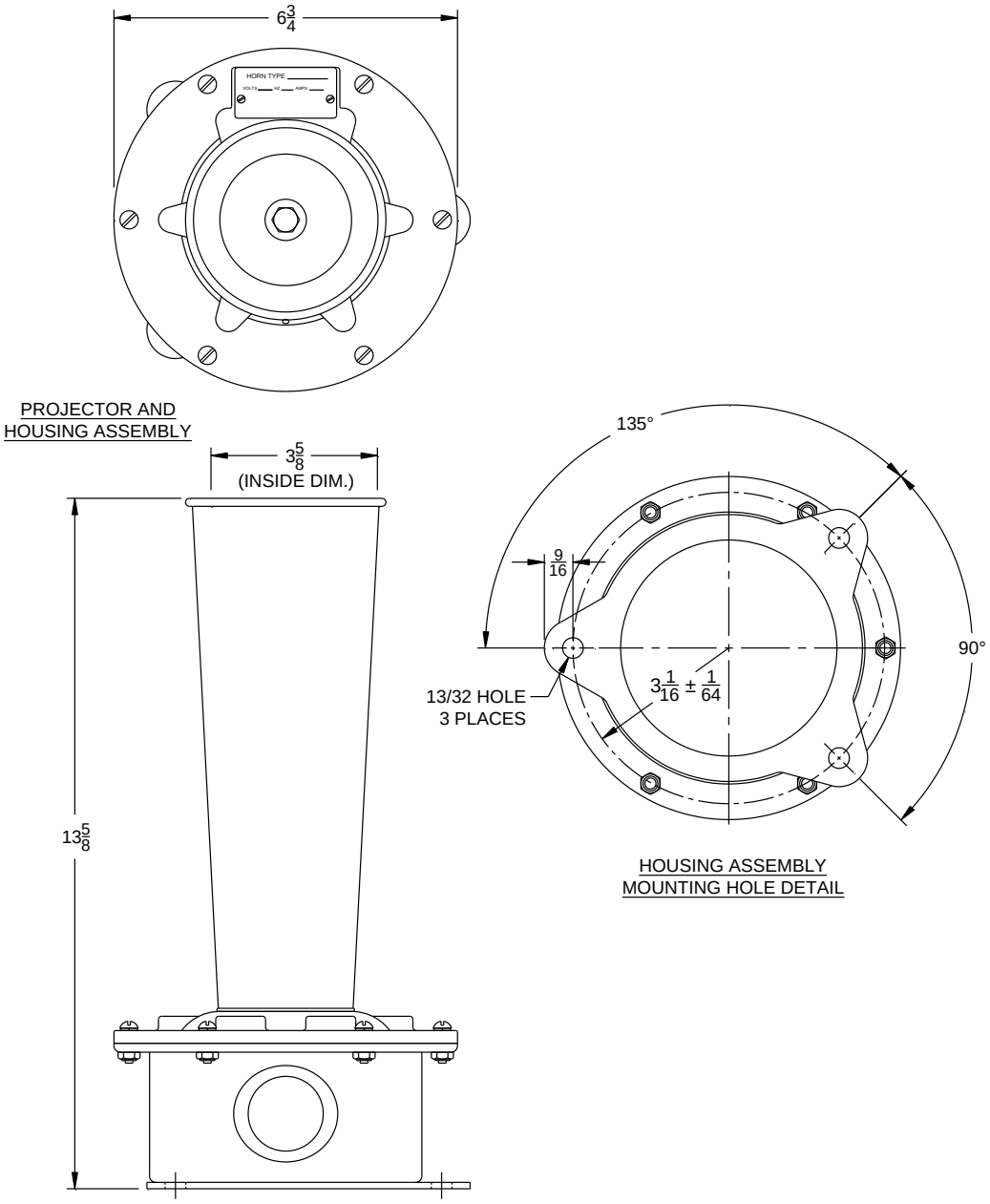


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. To obtain maximum sound output, horn should be bulkhead mounted using three spacer bosses ¾ O.D.X. ½ L with ⅜-16 tapped hole spacers to be welded to bulkhead on 6⅝ diameter bolt circle. Spacers furnished by installing activity.
3. All dimensions are in inches.

FIGURE 9. Watertight, vibratory horn (type IC/H1D4).

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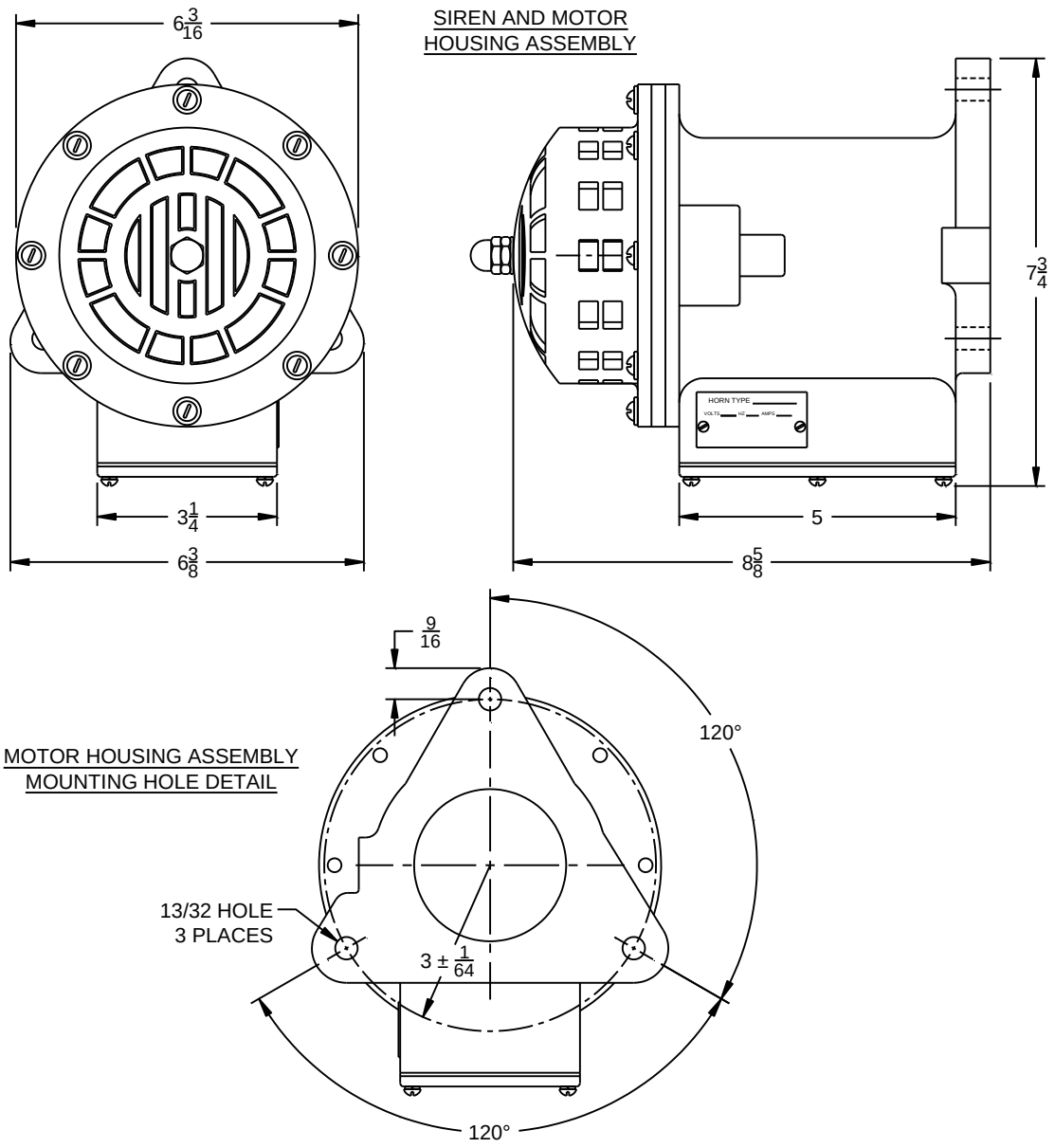


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. To obtain maximum sound output, horn should be bulkhead mounted using three spacer bosses ¾ O.D.X. ½ L with ⅜-16 tapped hole spacers to be welded to bulkhead on 6⅛ diameter bolt circle spacers furnished by installing activity.
3. All dimensions are in inches.

FIGURE 10. Watertight, vibratory horn (type IC/H1S4).

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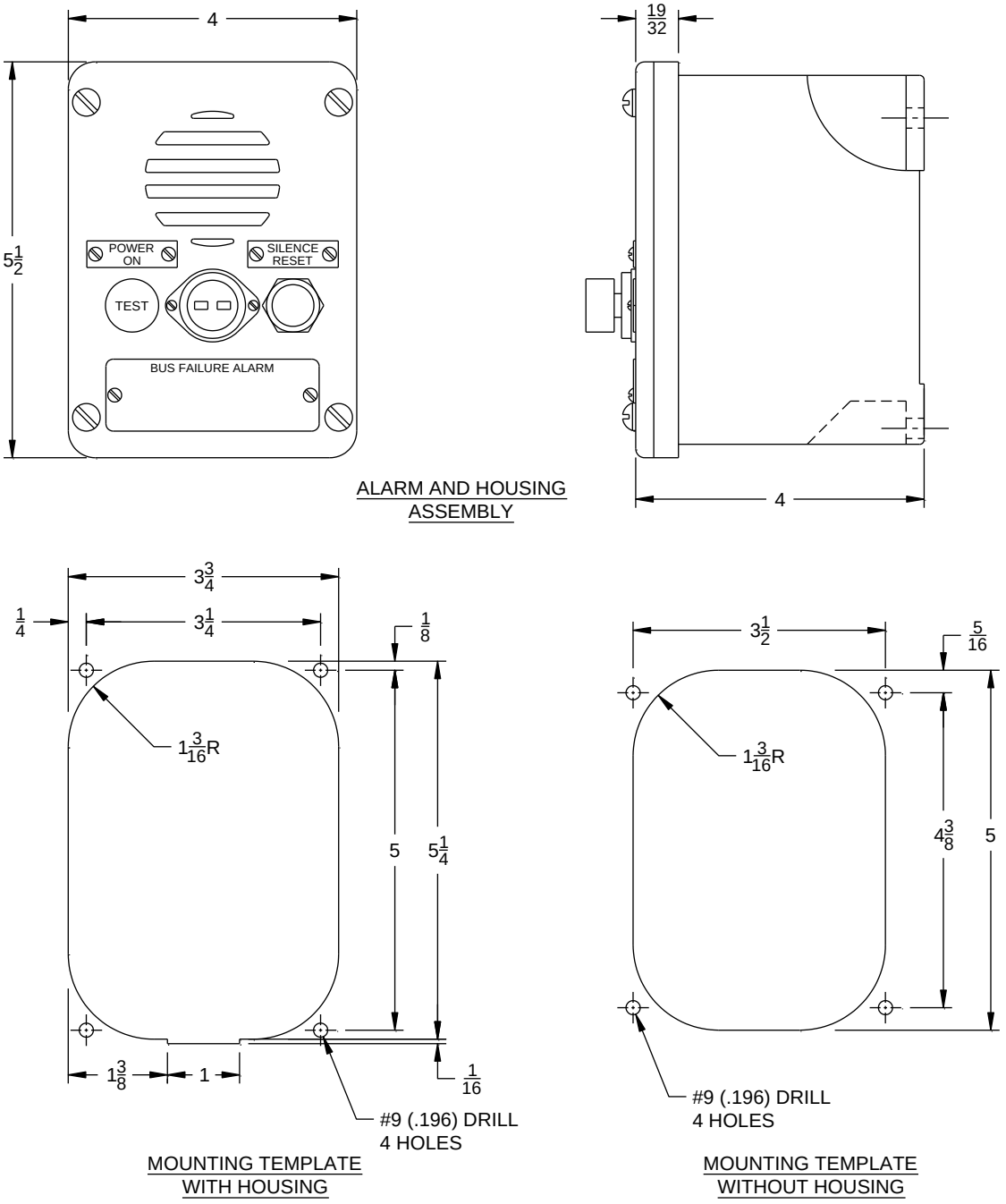


NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. All dimensions are in inches.

FIGURE 11. High and low pitch, drip proof sirens (type IC/S1D4, S2D4, S1S4, S2S4, S1S4CMS, and S2S4CMS).

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NOTES:

1. Unless otherwise specified herein, dimensions are maximum/not to exceed.
2. All dimensions are in inches.

FIGURE 12. Bus failure alarm signal (type IC/E1D1 and E1D2).

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6350-2016-006)

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